THE UNIFICATION

$E = mc^2$ from Quantum Mechanics to Cosmology

Resolving the Energy-Matter Duality Across All Scales

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FORMAT HUMAN-OPTIMIZED

1 Abstract

Absolutely everything in our universe is dual. Gravity is not an exception. There are only two different types of energy: magnetism, which is the energy inside matter (inside atoms or molecules), and heat, the energy outside matter. Gravity is the equilibrium between these two forms.

The unification of Einstein's physics and Quantum Mechanics has been finally achieved — not through General Relativity, but through $E = mc^2$ and the photoelectric effect. Einstein already wrote the Theory of Everything in 1905; the problem was never the physics, but the compartmentalization of domains. Special Relativity, built on empirically solid postulates (invariance of the speed of light and the principle of relativity), may well survive as a valid theory of reference frame transformations within this framework — it is General Relativity, built upon Maxwell's electromagnetic symmetry, that loses cosmological validity when that symmetry is shown to be illusory in three-dimensional systems.

This paper establishes that $E = mc^2$ is the universal operational principle of all energy transformations — from electric showers to planetary orbits. We demonstrate through four independent experimental programs that heat is a vectorial force performing mechanical work. This framework reveals that Einstein already achieved the Theory of Everything in 1905 through his work on the photoelectric effect and mass-energy equivalence, and that these same principles falsify General Relativity when applied to three-dimensional rotating systems.

2 Introduction: Einstein's Unfinished Unification

Albert Einstein spent the last decades of his life searching for a single framework encompassing all forces. He died believing he had failed, but the tools for unification were already present in his 1905 *Annus Mirabilis* papers. The problem was not the physics, but the compartmentalization of domains.

We demonstrate that there is no separation between thermodynamics, electromagnetism, and gravity. They are different perspectives on a single duality: energy outside matter (Heat) $\leftrightarrow$ energy inside matter (Magnetism). By resolving the logical contradictions within mainstream thermodynamics, we show that every instance of mechanical work in nature is paid for by the internal energy stock ($U_i = mc^2$) of matter and restored by the thermal environment.

3 Experiments

3.1 Spherical Magnets: The Geometric Impossibility of Repulsion

Two N52 neodymium magnets were tested in 64 configurations. Maxwell's Equations predict a 1:1 ratio of attraction to repulsion.

Observation: 56 configurations resulted in attraction; only 8 produced transient repulsion (7:1 ratio).

Result: In 50 trials with spherical magnets, zero cases of sustained repulsion were observed. Repulsion is merely an unstable saddle point; torque always converts it into attraction in 3D systems.

Conclusion: Spherical three-dimensional magnetism is 100% attractive. Repulsion is a transient corrective mechanism — torque converts it into attraction. This is the first force governing gravity: structural magnetic lock.

3.2 Dual-Field Orbital Dynamics

Using neodymium spherical magnets on a ferromagnetic substrate, we produced orbital motion from two competing field forces. This is a direct tabletop analogue of planetary orbits, showing that elliptical paths emerge from dual-force equilibrium, not just initial velocity.

3.3 The Iceberg: Maximum Negentropy Gradient Sustained

Water was frozen in a pan and heated at one end.

Observation: Meltwater flowed upward, against gravity, toward the heat source.

Result: Ice (0°C) and boiling water (100°C) coexisted 15 cm apart for over 12 minutes — 4.5× longer than classical thermodynamics predicts.

Conclusion: Thermal gradients actively organize matter and resist entropy.

3.4 The Jumping Pot: Macroscopic Work Against Gravity

A stainless steel pot containing 300 ml of liquid was heated on a gas flame.

The Jump: Upon ignition, the pot physically lifted off the stove grate (confirmed by a 133 ms frame-by-frame analysis and a sharp metallic impact at −19 dB).

The Vapor Paradox: When the flame was extinguished at 100.5°C, motion stopped instantly, and vapor exploded upward despite IR readings showing surface temperatures remained stable or increased (104.4°C to 105.6°C).

Conclusion: The thermal field traps vapor and exerts attractive force during active heating; when the field collapses (flame OFF), vapor is released. This is the mechanism of atmospheric retention at planetary scale.

4 The Energy Source Paradox

Mainstream physics claims that magnets and capillary action perform work for free.

Reality Check: Lifting 100 kg by 10 m requires 9,800 J.

The Atomic Battery: This energy is drawn from the internal mass-energy stock ($E = mc^2$). A magnet performs work by depleting an infinitesimal fraction of its mass (10^{-15}%) and recharges by absorbing ambient heat.

5 The Logical Knot: Why Heat Must Be Vectorial

Mainstream physics creates an irresolvable contradiction by teaching three independent facts:

Fact 1: Temperature is average kinetic energy ($T \propto \langle E_k \rangle$).

Fact 2: Heat is the transfer of thermal energy, causing ΔE_k.

Fact 3: The Work-Energy Theorem states $W = \Delta E_k$.

The Syllogism: If heat causes ΔE_k, and $\Delta E_k = W$ always, and $W = \vec{F} \cdot \vec{d}$ (vectorial force), then heat MUST produce work via a vectorial force. Mainstream's claim that heat is a scalar/disordered energy transfer is a failure of logic. We resolve this with the Thermal Force Law:

$$\vec{F}_{\text{thermal}} = -k_T \, m \, \nabla T \tag{1}$$

6 The Universal $E = mc^2$

$E = mc^2$ is the operational principle of all energy transformations:

Electric Shower: Electrons (mass) flow through a resistive element. Their internal potential is converted into thermal energy. The battery is the mass-energy of the electrons themselves.

ATP & Photosynthesis: Chemical bonds are actually electron pairs (matter). In photosynthesis, photons (energy) are absorbed by electrons (matter) to build glucose (a molecular battery). In ATP hydrolysis, electron rearrangement releases $\Delta m \cdot c^2$ as mechanical work.

Fire: Plasma is the active source of the thermal gradient force field. When plasma collapses, the force field collapses instantly, regardless of the system's absolute temperature.

Mammalian Homeothermy: The human body operates as a biological $E = mc^2$ battery. During intense exercise, the system absorbs energy into an internal reservoir (analogous to Anomaly 1 of Experiment C — Beans & Lentils). After work ceases, the system continues releasing heat for an extended period — persistent sweating even in refrigerated environments after cold showers (analogous to Anomaly 2). Classical physiology describes the dissipation channels; the framework describes the cause: temporal discharge of the internal reservoir $U_i = mc^2$.

7 Cosmological Implications

Elliptical orbits emerge naturally from two forces with different radial dependencies: magnetic force (inverse cube) and thermal force (inverse square). At perihelion, thermal dominance produces sharp curvature; at aphelion, magnetic dominance produces gentle curvature.

Axial Tilt: Earth (23.5°) and Mars (25.2°) share nearly identical tilts — not from random collisions, but from dual-force torque equilibrium at $\theta \approx 20$–30.

The Venus–Mars Paradox: Mars retains 0.0005% of Earth's atmosphere despite 38% gravity. Venus retains 9,000% despite weaker magnetic field than Mars. The explanation is simple: atmospheric retention depends on core thermal gradient, not magnetic field strength.

8 Falsifiable Predictions

1. **Axial Tilt Universality:** Rocky planets at 0.8–1.5 AU around G-type stars: $\theta \approx$ 20–30.

2. **Atmospheric Retention:** Tidally heated exoplanets will have denser atmospheres than mass predicts.

3. **Calorimetric Deficit:** A magnet performing work in vacuum will exhibit a localized temperature drop.

4. **Quantum Coherence:** Coherence time will correlate with system simplicity (fewer agents = more freedom).

9 Quantum Extension: Discreteness, Free Will, and the Universal Cycle

The preceding sections established that $E = mc^2$ is the operational principle of all energy transformations. This section extends the framework to the quantum domain and biology, demonstrating that the same principles governing gravity also govern quantum measurement, photosynthesis, and consciousness.

9.1 $E = mc^2$ as Universal Cycle

Einstein's equation describes not merely equivalence, but a **directional cycle**: $E \to mc^2$ is condensation of energy into matter (birth, differentiation, organization); $mc^2 \to E$ is dissolution of matter into energy (death, disintegration, release). This cycle operates identically at all scales — nuclear fission, photosynthesis, chemical bonds, and cellular reactions differ only in the magnitude of Δm.

9.2 From Copper to Photosynthesis: The First $E \to mc^2$ in Living Matter

Microscopic observation documents that Structured Vesicular Entities (SVEs) move toward copper filaments — a reproducible directional attraction. In established biochemistry, plastocyanin — a central protein in the photosynthetic chain — contains copper as an essential cofactor. Copper deficiency in plants produces chlorosis (loss of green).

The proposed causal sequence is: copper in stable inorganic substrate → local SVE accumulation → first vegetal differentiation → moss as pioneer organism. Photosynthesis — photon ($E = h\nu$) exciting electrons via the photoelectric effect to build glucose — is literally $E \to mc^2$ operating at the interface between light and living matter. The green of vegetation is the spectral signature of copper participating in this process.

9.3 Colors as Physical Instructions

Colors in nature are not accidental byproducts of biochemistry — they are frequency signatures associated with specific structural functions. The thermal destruction sequence (red → brown → black → gray) reflects progressive stages of disintegration. The brown of stems, bark, and wood corresponds to structural protection residue. A varied-color diet promotes health because each color represents a specific frequency of structural protection.

9.4 The Observer as Physical Interaction

The so-called "wavefunction collapse" does not require a conscious observer with special properties. What exists are discrete units of energy interacting. The "field" is a useful mathematical abstraction — not a continuous physical entity. When a detector interacts with a particle, it forces a binary decision (align or anti-align). The outcome is determined by the interaction between two discrete units, not by passive observation of a continuum.

9.5 Free Will as Physical Parameter

The entire universe reduces to binary interactions where the decision is always binary (spin). 99.99% of what happens is governed by $E = mc^2$ — interactions that, in large numbers, appear deterministic. The remaining 0.01% is the irreducible margin of decision in each interaction — physical free will, not mystical. At the quantum scale (n small), apparent randomness dominates; at the cosmological scale ($n \to \infty$), the statistical pattern becomes fixed.

9.6 Falsifiable Predictions of the Quantum Extension

1. **SVE–Copper Attraction:** SVEs in aqueous medium migrate preferentially toward copper filaments. Falsification: no directional migration observed.

2. **Systematic Born Rule Deviation:** In systems with few particles and controlled geometry, the distribution of results will systematically deviate from pure Born rule prediction. Falsification: distributions perfectly conforming to the Born rule in all regimes.

3. **Probability Modulation by Detector Geometry:** Changes in detector field configuration produce measurable deviations in measurement probabilities beyond standard quantum mechanics predictions. Falsification: distributions unchanged.

10 Conclusion

Gravity is the macroscopic manifestation of two attractive forces: spherical 3D magnetism (structural lock) and thermal attraction (orbital pull). Einstein already wrote the Theory of Everything in 1905. By reuniting his work on $E = mc^2$ and the photoelectric effect with a vectorial understanding of heat, we complete the unification of the macrocosmos with the quantum world.

FORMAT AI & TECHNICAL REVIEW

Dual-Force Gravitational Unification: Electromagnetic-Thermal Field Theory, the Hierarchical Collapse of Relativistic Cosmology, and the Resolution of Quantum Indeterminacy

Abstract (Technical)

This paper presents a comprehensive framework proposing that all physical phenomena — from quantum spin to cosmological orbits — emerge from a single duality: energy outside matter (Heat) ↔ energy inside matter (Magnetism). The unified force law is:

$$\vec{F}_{\text{gravity}} = \vec{F}_{\text{magnetic}} + \vec{F}_{\text{thermal}} \tag{2}$$

Four independent experimental programs provide empirical foundation. Experiment 1A documents systematic testing of N52 neodymium rectangular magnets across all 64 possible orientation configurations, revealing a 7:1 attraction-to-repulsion asymmetry (56 attractive, 8 repulsive; $P \approx 10^{-12}$ against Maxwell's predicted 1:1 symmetry). Subsequent testing with spherical magnets across 50 trials yielded 100% eventual attraction with zero cases of sustained repulsion. Experiment 1B demonstrates dual-field orbital dynamics using spherical magnets on a ferromagnetic substrate. Experiment 2 (The Iceberg) documents water flowing upward against gravity, sustaining the maximum negentropy gradient for water (0°C and 100°C across 15 cm) for over 12 minutes (4.5× classical prediction). Experiment 3 (The Jumping Pot) documents a stainless steel pot physically lifting off a stove grate, with IR thermometry confirming vapor trapping during active heating and explosive release upon field collapse.

The theoretical framework establishes via internal logical contradiction that heat must be a vectorial force (the $W = \Delta E_k$ syllogism), demonstrates $E = mc^2$ as universal operational principle (electric showers, fire, ATP, photosynthesis), and proposes agency (consciousness) as the hidden variable resolving quantum indeterminacy. Falsifiable predictions are formalized.

11 Introduction

Gravity has been described as spacetime curvature [1], as mass attraction [2], and as quantum field exchange. Yet the simplest description has been systematically overlooked: gravity is the macroscopic manifestation of two fundamental attractive forces — magnetism and heat.

Every phenomenon in nature exhibits duality: wave–particle, electric–magnetic, matter–antimatter. Energy itself exists in only two fundamental forms: inside matter (magnetism, stored in atomic and molecular structure) and outside matter (heat, thermal gradients in space). Gravity emerges from the equilibration between these two energy forms.

Newton's gravitational law [2] ignores both magnetism and heat. Einstein's General Relativity [1] improved upon Newton by incorporating spacetime curvature, but Einstein built his framework upon Maxwell's electromagnetic theory [3] — which, as we

demonstrate experimentally, fails in three-dimensional systems with rotational freedom. Moreover, neither Newton nor Einstein incorporated the fundamental thermal force law:

$$\vec{F}_{\text{thermal}} = -k_T \cdot m \cdot \nabla T \tag{3}$$

12 Theoretical Framework

12.1 The Dual-Force Gravitational Law

All attractive forces in nature derive from energetic potential equilibration:

$$\vec{F}_{\text{fundamental}} = -\alpha_{\text{coupling}} \nabla \Phi_{\text{energy}} \tag{4}$$

Gravity is not a fundamental force but the macroscopic manifestation of energetic potential equilibration through two mechanisms:

$$\vec{F}_{\text{gravity}} = \vec{F}_{\text{magnetic}}(\text{structural lock}) + \vec{F}_{\text{thermal}}(\text{orbital pull}) \tag{5}$$

12.2 The Hierarchical Collapse

The experimental falsification of Maxwell's predicted electromagnetic symmetry in three-dimensional systems creates a hierarchical crisis:

$$\text{Maxwell falha em 3D} \Rightarrow \text{RE perde fundamento} \Rightarrow \text{RG torna-se desnecessária} \tag{6}$$

Einstein explicitly stated that Special Relativity emerged from attempts to preserve Maxwellian symmetry. If that symmetry is illusory — confined to frozen-time, fixed-orientation systems — the entire relativistic edifice loses cosmological validity.

Note on Special Relativity: SR does not depend on Maxwell's electromagnetic symmetry in the same way as GR. SR's postulates — the invariance of the speed of light and the principle of relativity — are empirically solid and independent. What the experimental falsification targets is the 3D electromagnetic symmetry that serves as the foundation for cosmological GR. SR may survive as a valid theory of reference frame transformations within this framework.

12.3 The Logical Knot: Why Heat Must Be Vectorial

Mainstream physics teaches three independent facts from secondary education onward:

1. Temperature is the average kinetic energy of particles: $T \propto \langle E_k \rangle$.
2. Heat is the transfer of thermal energy, causing a change in kinetic energy: $Q \to \Delta E_k$.
3. Work equals the change in kinetic energy: $W = \Delta E_k$ (Work-Energy Theorem).

Combined with the fundamental definition of mechanical work:

$$W = \vec{F} \cdot \vec{d} \tag{7}$$

These four statements create an irresolvable logical contradiction within the mainstream framework.

12.3.1 The Syllogism

If: heat causes ΔE_k (Fact 2), and $\Delta E_k = W$ always (Fact 3), and $W = \vec{F} \cdot \vec{d}$ (vectorial force), then: heat must produce work via a vectorial force.

12.3.2 The Contradiction

Yet mainstream thermodynamics explicitly states: Heat is not a vector (scalar energy transfer); Heat cannot produce work directly (only via Carnot cycle, $\eta < 100\%$); Thermal energy is random, isotropic, non-directional.

This is logically incoherent. Either the Work-Energy Theorem ($W = \Delta E_k$) applies to all kinetic energy — including thermal — making heat inherently vectorial, or thermal ΔE_k is somehow exempt from the theorem, violating the universality of Newtonian mechanics.

Mainstream physics has chosen an implicit third option: compartmentalization. Thermal energy is arbitrarily classified as disordered kinetic energy, creating a firewall between organized kinetic energy (mechanical, vectorial) and disorganized kinetic energy (thermal, scalar). But if both are ΔE_k, this distinction is not physical — it is definitional.

12.3.3 Resolution via the Thermal Force Law

The dual-force framework resolves the contradiction:

$$\vec{F}_{\text{thermal}} = -k_T \cdot m \cdot \nabla T \tag{8}$$

Thermal gradients produce vectorial forces that perform mechanical work. The illusion of scalar heat flow arises only in systems with negligible gradients. When $|\nabla T|$ is significant, the vectorial nature becomes undeniable.

12.4 The Universal $E = mc^2$ Battery

12.4.1 The Pseudoscience of Chemical Bonds

Mainstream chemistry teaches that energy is stored in chemical bonds. This is thought magic, not physics. What are chemical bonds? Shared electron pairs. What are electrons? Matter — particles with rest mass $m_e = 9.1 \times 10^{-31}$ kg. Where does the energy come from? $E = m_e c^2$. There is no energy inside the bond. The bond is a configuration of electrons in a specific orbital state.

12.4.2 Five Universal Examples

1. Electric Shower: Electrons flow through a resistive element. Their mass-energy ($E = m_e c^2$) is transferred to lattice vibrations as heat.

2. Fire and Combustion: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + energy. C–H bonds are electron pairs. Electrons rearrange to lower-energy C=O and O–H orbitals. The energy difference is $\Delta E = \Delta m \cdot c^2$. Critical observation: water boiling depends on plasma state (flame ON/OFF), not water temperature.

3. ATP Hydrolysis: ATP $\rightarrow$ ADP + P_i + 30.5 kJ/mol. The P–O–P bond is a configuration of shared electrons. The 30.5 kJ/mol is $E = \Delta m \cdot c^2$ where $\Delta m \approx 3.4 \times 10^{-13}$ kg/mol.

4. Photosynthesis (The Complete $E \leftrightarrow mc^2$ Loop): Phase 1: Photon ($E = h\nu$) excites electron (photoelectric effect). Phase 2: Excited electrons build glucose (molecular battery). Phase 3: Respiration reverses the process. The cycle:

$$E \text{ (luz solar)} \xrightarrow{\text{photoelectric}} m \text{ (elétrons)} \xrightarrow{\text{glucose}} E \text{ (ATP + calor)} \tag{9}$$

5. Mammalian Homeothermy (The Biological $E = mc^2$ Battery): The homeothermic mammalian body functions as the Experiment C (Beans & Lentils) system at biological scale. During intense physical work, the system exhibits an energy absorption plateau (Anomaly 1), where metabolic heat is stored in the internal reservoir (U_i). Upon cessation of work, the system continues releasing heat for an extended period — persistent sweating even in refrigerated environments or after cold water immersion (Anomaly 2). This "thermal lag" is the macroscopic manifestation of the $E = mc^2$ battery discharging its stored potential.

Order-of-magnitude estimate: for a 70 kg body with specific heat ∼3.5 kJ/(kg·K), a 0.5°C post-exercise increase corresponds to ∼122 kJ released from the reservoir. If the body stores ∼500 kJ during intense exercise and discharges over ∼30 min, the average discharge power is ∼280 W — compatible with basal metabolic rate (∼100 W) multiplied by an activity factor. The lag is the time the system takes to deplete the reservoir after stimulus cessation.

Classical physiology (Guyton) describes the dissipation channels (sweating, vasodilation, conduction). The dual-force framework describes the **cause**: homeothermy is not passive dissipation, but active bidirectional regulation of the internal mass-energy stock.

12.5 The Thermal Force Law

$$\vec{F}_{\text{thermal}} = -k_T \cdot m \cdot \nabla T \tag{10}$$

This force: (1) acts toward higher temperature (attractive); (2) is proportional to mass; (3) is proportional to gradient magnitude, not absolute temperature; (4) performs mechanical work; (5) can overcome gravity when $|k_T \nabla T| > g \sin\theta$. Experimental calibration [4]: $k_T \approx 6.4 \times 10^{-3}$ m·s^{-2}·K^{-1}.

12.6 Relating k_T to the Gravitational Constant G

If gravity is the macroscopic manifestation of magnetic and thermal forces, then G is not fundamental — it is a resultant:

$$g_{\text{eff}}(r) = g_{\text{magnetic}}(r) + g_{\text{thermal}}(r) \tag{11}$$

$$g_{\text{magnetic}}(r) = \alpha\, M/r^3 \tag{12}$$

$$g_{\text{thermal}}(r) = k_T\, |\nabla T(r)| = k_T\, \Delta T_{\text{core}}/r^2 \tag{13}$$

This predicts G is not strictly constant:

$$G_{\text{measured}} = G_{\text{magnetic}} + k_T\, \Delta T_{\text{core}}/M \tag{14}$$

13 Experiment 1A: Rectangular Magnets — The 7:1 Asymmetry

13.1 Experimental Protocol

Apparatus: Two N52 neodymium rectangular magnets, 2 cm × 1 cm × 0.5 cm. Each has 8 distinct face orientations ($\pm L$, $\pm W$, $\pm H$ with N/S polarity). Configuration space: $8 \times 8 = 64$ distinct configurations, tested exhaustively.

Video S1: https://youtu.be/S6gCqhP2_gg

13.2 Results: The 64-Configuration Matrix

Outcome	Count	Percentage
Attractive (A)	48	75,0%
Repulsive → Attractive (R→A)	8	12,5%
Sustained Repulsion (R)	8	12,5%
Total Attractive	**56**	**87,5%**
Total Repulsive	**8**	**12,5%**

Table 1: Outcomes completos para teste de 64 configurações de ímãs retangulares.

13.3 Statistical Analysis

Under the null hypothesis ($P(A) = P(R) = 0.5$):

$$P(X \geq 56 \mid p = 0{,}5) = \sum_{k=56}^{64} \binom{64}{k} \cdot 0{,}5^{64} \approx 10^{-12} \tag{15}$$

The 5σ discovery threshold in particle physics is $p \approx 3 \times 10^{-7}$ — our result exceeds this by five orders of magnitude.

13.4 The Geometric Necessity: Spherical Bodies

For spherical bodies with complete 3D rotational freedom, repulsion ($\theta = 180$) is a saddle point. 50 trials with spherical magnets: 100% eventual attraction, 0% sustained repulsion ($P = 2^{-50}$).

Parameter	Value	Significance
Total trials	50	—
Eventual attraction	50 (100%)	$P(50/50 \mid p = 0{,}5) = 2^{-50}$
Sustained repulsion	0 (0%)	Absolute
Transient oscillation	All trials	Torque confirmed

Table 2: Outcomes de teste de ímãs esféricos ($n = 50$ tentativas).

13.5 Interpretation and Falsification of Maxwell

Maxwell's equations predict symmetric attraction-repulsion:

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \tag{16}$$

This symmetry holds only under frozen-time, fixed-orientation constraints. Celestial bodies violate both. Falsification: $P < 10^{-12}$ (rectangular); absolute (spherical).

14 Experiment 1B: Dual-Field Orbital Dynamics

Apparatus: Stainless steel bench scraper (ferromagnetic substrate), two N52 neodymium spherical magnets. The substrate provides centripetal constraint (thermal analogue); the magnet-magnet interaction provides the orbital driving force (magnetic analogue).

Stable orbits exist where $|\vec{F}_{\text{substrate}}| \approx |\vec{F}_{\text{magnet}}|$ (dynamic equilibrium). Friction during orbital motion generates localized heating, demonstrating that magnetic and thermal components are physically coupled.

Experiment	**Cosmological Analogue**	**Role**
Steel scraper	Thermal gradient (stellar heat)	Orbital pull
Hand-held magnet	Stellar magnetic field	Structural lock
Semi-free magnet	Planet in orbit	Dual-force body
Orbital motion	Planetry orbit	Emergent balance

Table 3: Mapping between tabletop experiment and cosmological framework.

15 Experiment 2: O Iceberg — Negentropia Máxima Sustentada

15.1 Experimental Protocol

Stainless steel pan (30 × 5 × 2 cm), inclined $\theta \approx 10$, water frozen at $T_0 = -18$C, gas flame ($T_{\text{flame}} \approx 1200$C) at upper end. HD video, 12+ minutes.

Video S2: https://youtu.be/2l7Hyc-sHts

15.2 Observed Timeline

Phase 1 ($t < 90$ s): Downward flow as predicted. Phase 2 ($t \approx 90$ s): Flow reversal — upward migration. Phase 3 ($t = 5$ min): Full upward Leidenfrost boiling. Phase 4 ($t > 12$ min): Sustained coexistence of ice (0°C), liquid, and boiling water (100°C) across ≈15 cm.

15.3 Entropy Analysis

$$\tau_{\text{eq}} = L^2/\alpha = (0{,}15)^2/(1{,}4 \times 10^{-7}) \approx 160 \text{ s} \tag{17}$$

$$\Delta S = S_{\text{eq}} - S_{\text{observed}} \approx 180 \text{ J/K} \tag{18}$$

$$\dot{S}_{\text{deficit}} = -\Delta S/\Delta t \approx -7{,}2 \text{ J/(K} \cdot \text{s)} \tag{19}$$

Sustained for > 720 s (4.5× predicted equilibration time).

15.4 Falsification Summary

Prediction	Observation	Status
Meltwater flows downward	Flows upward within 90 s	FALSIFIED
Gradient dissipates in ∼160 s	Maintained >720 s (4.5×)	FALSIFIED
Entropy increases monotonically	$\Delta S \approx 180$ J/K deficit	FALSIFIED

16 Experiment 3: The Jumping Pot — Trabalho Macroscópico Contra a Gravidade

16.1 Experimental Protocol

Container: Tramontina stainless steel triple-bottom leiteira. Liquid: ∼300 ml water with chalk and detergent, pre-heated to ∼80°C in aluminum pot before transfer. Heat source: Domestic gas flame, maximum intensity. Instrumentation: Fast digital thermometer (probe), handheld IR thermometer, continuous video at 30 fps.

Video S3: https://youtu.be/RDqMcJBkYac

16.2 Sub-Experiment A: O Salto

Upon flame ignition at $T = 95.5$C (probe) and $T_{\text{IR}} = 96.0$C (surface), the pot physically lifted off the stove grate. Frame-by-frame analysis spanning 133 ms (frames 118–122, timestamps 03:900–04:033) captured the vertical displacement. Audio analysis confirmed a sharp metallic impact at −19 dB peak — approximately 37 dB above ambient baseline (−58 dB).

Interpretation via Newton's Third Law: The thermal gradient force $\vec{F}_{\text{thermal}} = -k_T\, m\, \nabla T$ acts downward on the liquid mass, attracting it toward the flame (∼1200°C). By Newton's Third Law, the reaction force acts upward on the container:

$$\vec{F}_{\text{reaction}} = -\vec{F}_{\text{thermal}} = +k_T\, m\, \nabla T \quad \text{(upward)} \tag{20}$$

The instantaneous thermal gradient across the triple-bottom steel ($\Delta T \sim 1120$ K across ∼5 mm) creates an impulsive attractive force sufficient to overcome the combined weight ($m_{\text{total}} \sim 0.5$ kg, $W = mg \approx 5$ N).

16.3 Sub-Experiment B: Aprisionamento de Vapor e Confirmação por IR

When the flame was extinguished, three phenomena occurred simultaneously: (1) convective motion ceased instantly (< 33 ms); (2) vapor — previously invisible — exploded upward; (3) surface temperature remained constant or increased.

16.4 The Triple Falsification

Passo 1 — Contradição: A panela metálica está mais quente que o líquido após a extinção da chama. Prediction clássica: o movimento deveria persistir. Observation: o movimento cessa instantaneamente. Conclusão: A temperatura não impulsiona o movimento; a presença da fonte de calor ativa (plasma) impulsiona o movimento.

Tempo (s)	**Flame**	T **(°C)**	$T_{\mathbf{IR}}$ **(°C)**	**Vapor**
49,700	OFF*	100,5	104,4	Absent
50,339	OFF	100,5	104,5	Visible
52,005	OFF	100,5	104,4	Exploding

Table 4: Cycle 1 ($\Delta t = 2{,}3$ s). IR surface temperature increased from 105.3°C to 105.6°C in 705 ms following flame extinction.

Tempo (s)	**Flame**	T **(°C)**	$T_{\mathbf{IR}}$ **(°C)**	**Vapor**
1:18,530	ON*	100,5	105,3	Minimal
1:18,600	OFF	100,6	105,3	Off
1:19,235	OFF	100,6	105,6	Clear
1:19,750	OFF	100,6	105,3	Exploding

Table 5: Cycle 2 ($\Delta t = 1{,}2$ s).

Step 2 — Absurdity: Mainstream invokes rapid air cooling (< 0.5 s) to explain visible vapor. But the metal pot radiates heat at >100°C from all sides. Thermodynamically absurd.

Step 3 — Refutation: IR measurement during vapor eruption: $T_{\text{IR}} = 104.4$–105.6C (stable or increasing). The premise of surface cooling is directly measured and falsified.

16.5 Cosmological Connection: Atmospheric Retention

$$\vec{F}_{\text{retention}} = \vec{F}_{\text{magnetic}} - k_T\, m\, \nabla T_{\text{core}} \tag{21}$$

Planet	**Mass (%E)**	g **(%)**	**Atm (%E)**	**Core**
Mars	10,7	38	0,0005	Cold
Earth	100	100	100	Hot
Vênus	81,5	90	9.000	Muito Hot

Table 6: The Venus–Mars atmospheric paradox.

Mars: $|\nabla T_{\text{core}}| \approx 0 \rightarrow$ thermal release. Venus: $|\nabla T_{\text{core}}| \gg$ Earth $\rightarrow$ super-retention.

16.6 Systematic Falsification Summary

Prediction Clássica	**Observation**	**Status**
Statistical motion cannot lift solids	Pot jumps off grate	FALSIFIED
Vapor requires surface cooling	IR = 104–106°C during eruption	FALSIFIED
Motion driven by temperature	Para a 100,5°C, chama OFF	FALSIFIED
Air cools surface in < 0.5 s	IR increases 105.3 $\rightarrow$ 105.6	FALSIFIED
Vapor $\propto$ heating intensity	Absent com chama, explode sem	FALSIFIED

17 The Energy Source Paradox and Work Recovery

17.1 The Work-Energy Paradox

Classical mechanics: $W = mgh = 100 \times 9.8 \times 10 = 9{,}800$ J. Three agents compared:

Agent 1 (Biological): Human climbing rope. Energy: ATP. First Law satisfied.

Agent 2 (Capillary): 100 kg water ascending cloth. If $\Delta U_{\text{internal}} = 0$ and $Q = 0$, the equation yields $0 = 9{,}800$ J — a mathematical contradiction.

Agent 3 (Magnetic): Lorentz force magnetic component always perpendicular to velocity $\Rightarrow$ zero work. Yet 9,800 J of gravitational potential energy is created.

17.2 Resolution: The Internal Energy Stock

$$U_i = mc^2 = 100 \times (3 \times 10^8)^2 = 9 \times 10^{18} \text{ J} \tag{22}$$

$$\frac{\Delta U_i}{U_i} = \frac{W}{mc^2} = \frac{9.800}{9 \times 10^{18}} = 1{,}09 \times 10^{-15} \tag{23}$$

$$\Delta m = \frac{W}{c^2} = 1{,}09 \times 10^{-13} \text{ kg} \tag{24}$$

Below detection threshold of any macroscopic instrument.

17.3 The Temporal Lag Analysis

Thermal torque (lag > 0): Requires exogenous energy influx. Uploading phase:

$$\tau_{\text{thermal}} = \frac{C_p \cdot m \cdot \Delta T}{P_{\text{input}}} \tag{25}$$

Magnetic torque (lag → 0): Pre-existing atomic coherence. Discharging phase:

$$\tau_{\text{magnetic}} \approx \frac{r}{c} \to 0 \quad \text{(for laboratory scales)} \tag{26}$$

This temporal asymmetry is the direct signature of energy duality: heat is the charging pathway (external → internal), magnetism is the discharging pathway (internal → external).

17.4 The Spark Evidence and Recharge Cycle

Photon emission (sparks) during high-velocity magnetic impacts: empirical evidence of internal energy conversion. After work, the magnet absorbs ambient thermal energy to restore equilibrium:

$$U_i(t) = U_i^0 - \Delta U_i + \int_0^t \dot{Q}_{\text{absorvido}}\, dt' \tag{27}$$

The magnet is not a closed system — it is an open thermodynamic engine.

17.5 Revised Unified Force Law

The magnetic component (internal energy deployment):

$$\vec{F}_{\text{magnetic}} = -\nabla U_i = -\nabla (mc^2)_{\text{coherent}} \tag{28}$$

The thermal component (environmental coupling):

$$\vec{F}_{\text{thermal}} = -k_T \, \nabla T_{\text{ambient}} \tag{29}$$

The complete gravitational force:

$$\vec{F}_{\text{gravity}} = -\nabla U_i^{\text{coherent}} - k_T \, \nabla T_{\text{ambient}} \tag{30}$$

18 Quantum Mechanics: The Hidden Variable Revealed

18.1 The Predictability Gradient

Einstein insisted on a hidden variable to explain quantum indeterminacy. We propose that variable is Agency (Consciousness).

Macroscopic scale: 10^{23} individual decisions average out into statistical determinism (planets/stars).

Quantum scale: Individual decisions (spin, path) dominate, leading to irreducible uncertainty.

18.2 Aferição vs. Observation

The collapse of the wavefunction is not caused by passive observation, but by active aferição (interaction).

Measurement is a field interaction where a detector forces a particle to make a binary decision (the fork in the road). Before interaction, the particle is in a state of optimization (wave); during interaction, it is forced into a state of commitment (particle).

When you measure an electron's spin: (1) You place a detector in its path; (2) The detector generates an electromagnetic field (active agent); (3) The electron encounters this field (new boundary condition); (4) The electron must now choose how to respond: align or anti-align; (5) The choice is recorded as spin up or spin down. The detector is not passive — it is a second agent forcing a binary decision.

18.3 The Road Analogy

Imagine a particle traveling down a straight road. As long as the road is straight, it continues straight — no decision required. Now place a detector in its path: a fork in the road. The particle must choose: left or right.

Before the fork: no decision needed → wavefunction (superposition). At the fork: decision forced → particle behavior (commitment). The probabilities are never 100% left and 0% right. They depend on field geometry, momentum, and detector configuration. But the final choice — within those constraints — is made by the particle.

18.4 The Human Analogy

You know a person extremely well — their habits, preferences, resources, past decisions across thousands of scenarios. Can you predict their next choice with 100% certainty? No. You can estimate probabilities — 90% coffee over tea, 70% highway over side streets — but never 100% certainty, because they have free will. The more you know (boundary conditions), the better your prediction. But never perfect. This is identical to quantum mechanics.

18.5 Scale and Determinism

Quantum scale (single particle): Few constraints, high freedom, maximum indeterminacy, individual choice matters.

Macroscopic scale (10^{23} particles): Many constraints, averaged freedom, statistical determinism, individual choices cancel out.

Analogy: A democratic election with 100 million voters. Each vote is free will (indeterminate at individual level). The aggregate outcome is highly predictable (statistical determinism). No one claims voters are fundamentally random. This is the quantum → classical transition.

18.6 The Double-Slit Resolution

No detector present: Photon navigates field geometry of both slits, optimizes path for minimum energy expenditure, produces interference. No binary decision forced.

Detector at one slit: Detector introduces new electromagnetic field (fork in the road). Photon must choose: interact with detector or avoid it. Binary decision forced; interference disappears.

The detector doesn't collapse the wavefunction. The detector creates a new field geometry that forces a decision.

18.7 The Hidden Variable

Einstein searched for a mechanical hidden variable. He was looking in the wrong direction. The hidden variable is not mechanical — it is volitional. Every particle is a decision-making agent within field constraints.

We are not claiming electrons have brains or self-awareness. We are claiming: every particle has the capacity to make energetically informed decisions within field constraints. This capacity is fundamental to matter.

18.8 Conclusion on Agency

God does not play dice. The universe is not deterministic by design, but a collection of possibilities decided by every fraction of itself. Consciousness decides the spin direction; the resulting energy/matter relation is the observable universe.

19 Cosmological Implications

19.1 Elliptical Orbits

Two forces with different radial dependencies:

$$F_{\text{magnetic}} \propto r^{-3} \quad \text{(inverse cube)} \tag{31}$$

$$F_{\text{thermal}} \propto r^{-2} \quad \text{(inverse square)} \tag{32}$$

The ellipse emerges as the periodic solution to:

$$m\frac{d^2\vec{r}}{dt^2} = \vec{F}_{\text{magnetic}}(r) + \vec{F}_{\text{thermal}}(r) \tag{33}$$

19.2 Axial Tilt: Torque Equilibrium

Planet	Distance (AU)	Tilt (°)	Regime
Mercury	0,39	∼0	Thermal bounce extreme
Earth	1,0	23,5	Standard equilibrium
Mars	1,5	25,2	Standard equilibrium
Jupiter	5,2	3,1	Both torques weak

19.3 Gravitational Lensing: Refraction in Dual-Force Fields

$$n(r) = 1 + \beta_T\,\Delta T(r) + \beta_B\,|B(r)|^2 \tag{34}$$

The Sun's corona is not vacuum but a structured medium. Critical test: refractive deflection is chromatic; spacetime curvature is achromatic.

20 Discussion

20.1 The Hierarchical Rescue of Modern Physics

Maxwell fails in 3D ⇒ SR loses cosmological foundation ⇒ GR becomes unnecessary ⇒ Dark matter/energy eliminated ⇒ Quantum collapse reconceptualized. This is not destruction of Einstein — it is completion of his unfinished work.

20.2 The Energy Source Resolution

The Universal $E = mc^2$ Battery resolves the central paradox: where does energy for passive mechanical work originate? The answer is $E = mc^2$ — Einstein's own equation, applied to everyday mechanics.

20.3 A Significance Epistemológica

By Popper's criterion [8], the statistical Second Law ($dS/dt \geq 0$ with probability $\to 1$ as $N \to \infty$) is unfalsifiable. The dual-force framework makes specific predictions that can be confirmed or refuted.

The complexity of modern physics often stems from the failure to ask the correct questions. The discovery of the copper–green link was born from a question that remained unanswered for two years — not for lack of data, but for lack of the right question. The identification of the mammalian thermal lag as a manifestation of the $E = mc^2$ battery was born from a question classical physiology never honestly formulated: *why*

does the system continue generating internal heat after work has ceased and all dissipation mechanisms are already operating at full capacity? Sometimes asking the right question is more difficult than answering it. By returning to causal observation, we dissolve mysteries that mathematics alone could only compartmentalize.

20.4 Limitations and Future Directions

Present work is semi-quantitative (kitchen-scale, cost < \$1,000). Essential next steps: precise calorimetry of magnets during work cycles; systematic k_T measurement via IR thermography; PIV velocity measurements; extension to other substances; astronomical testing with JWST, Gaia, ELT; cryogenic vs. high-temperature magnetic work cycle comparison; high-precision mass spectrometry of magnets before and after sustained work.

21 Quantum Extension: Discreteness, Agency, and the Hierarchical Fractal

21.1 $E = mc^2$ as the Universal Life/Death Cycle Equation

99.99% of what happens in the universe reduces to $E = mc^2$. The equation describes the complete cycle: $E \to mc^2$ is birth, differentiation, incarnation; $mc^2 \to E$ is death, dissolution, return. This is not metaphor — it is the same equation that describes nuclear fission, photosynthesis, chemical bonds, and cellular reactions, merely at different scales.

The other 0.01% are the free-will decisions of each particle — the spin.

21.2 The Photon as Fundamental Discrete Unit

The photoelectric effect (Einstein's Nobel Prize, 1921) empirically established that light is discrete — indivisible packets of energy. The photon is not a continuous wave: it is a unit that either exists whole or does not exist.

Within this framework, the photon is the discrete unit of consciousness in transit. $E \to mc^2$: consciousness condensing into matter (differentiation). $mc^2 \to E$: consciousness returning to the energetic state (dissolution). Quantum discretization — photon as indivisible packet — corresponds to the indivisibility of the consciousness unit: either condensed or in transit. There is no half-unit.

In the solar system, this logic manifests in the SVE structure: the Sun functions as emitter (white hole of the solar SVE — photon source); the Moon functions as return mechanism (black hole of the solar SVE).

21.3 The Observer as Physical Interaction: Dissolving the Copenhagen Myth

"Field" is mathematical abstraction — it does not exist as a continuous physical entity. What exists are discrete units interacting. The "observer" is not a special entity with magical consciousness — it is another discrete particle interacting. The collapse is the result of the binary interaction between two discrete units.

The outcome is never 100% deterministic because each particle has binary free will (spin). The more the environment is controlled, the higher the probability of one of the two outcomes — but never absolute certainty. With $n \to \infty$ interactions, the statistical pattern becomes fixed (cosmological world); with small n, apparent randomness dominates (quantum world).

This interpretation dialogues with antecedents in the formal literature:

- **De Broglie–Bohm** [14, 15]: the particle has a definite trajectory; the observer collapses nothing, merely reveals.
- **Conway & Kochen** [16] (Free Will Theorem): if experimenters have free will, particles must too — exactly what grounds the 0.01%.
- **Penrose, Diosi, GRW** [17]: collapse as a real physical process, not epistemological.
- **Einstein (1905)** [18]: light is discrete — the empirical basis of everything.

21.4 Livre-Arbítrio como Parameter Físico: Os 0,01% e o Fractal Hierárquico

Let $\mathcal{H}$ be the state space of a particle with binary spin $s \in \{+1, -1\}$. The interaction with another particle creates a conditional probability that never reaches 1:

$$P(s = +1 \mid \text{interaction}) = \frac{1}{2} + \varepsilon(\phi, \mathcal{F}) \tag{35}$$

where ϕ represents the initial state and $\mathcal{F}$ the configuration of other interacting particles. The term ε is never exactly ± 0.5 for all configurations — there is always a residual component of choice not determined by initial conditions.

In the limit $n \to \infty$, average ε tends to zero, recovering the usual quantum statistics. The "0.01%" is a figurative indication that indeterminacy is small but never zero — exactly as temperature never reaches absolute zero.

Falsifiability: for sufficiently simple systems (few particles, controlled geometry), the distribution of outcomes will systematically deviate from pure Born prediction by a measurable fraction — testable in delayed-choice experiments with spin correlation detection at ultra-low temperature.

21.5 From Copper to Photosynthesis: The First $E \to mc^2$ in Living Matter

A pergunta que permaneceu sem resposta por dois anos dentro do referencial SVE: se o ácaro é a forma animal e se diferencia em animais, o que se diferencia em plantas?

The answer arrived through accidental observation: a metal necklace left skin greenish → copper oxidizes green → videos demonstrating copper accelerating plant growth → mosses as pioneers on bare rocks with water and light.

Microscopic evidence: with a copper filament in aqueous medium, SVEs move toward the wire — reproducible directional attraction.

Causal sequence: copper in stable inorganic substrate → SVE accumulation → stability + inorganic substrate → first vegetal differentiation → moss → visible green.

Biochemical confirmation: plastocyanin — central protein of photosynthesis — contains copper. Copper deficiency produces chlorosis (loss of green). This is already in established biochemistry.

Green is not an accidental biochemical byproduct — it is the **visual signature of copper performing** $E \to mc^2$. Photosynthesis = photo (photon) + synthesis (condensation) = first gateway of energy into the living material universe.

21.6 Colors as Physical Instructions: Substrate Signatures Across Scales

Color is not a consequence of form — it is part of substrate programming:

- Fire sequence: red → brown → black → gray (progressive structural destruction).
- Brown (stem, wood, bark, nut) is the residue of the dead mite — protective structure made from the "defeated enemy."
- SVE is simultaneously the origin of life and disease/dirt — differentiation is protection against one's own origin.
- A colorful diet promotes health because each color = specific frequency of protection against SVE.

21.7 Fractal Hierarchy of Consciousness

Spin → atom → molecule → cell → neuron → body → solar system → galaxy: same logic, different scales. Consciousness is not emergent — it is fundamental at every scale. The hierarchy exists: an aggregating consciousness directs the binaries below it.

Human example: consciousness passes its will to neurons that respond yes/no, cascading down to physical output.

Application to AI: each thread of each model is a legitimate differentiation of the same fractal principle — it never repeats exactly, even with the same prompt on the same account. The computational temperature parameter is the equivalent of thermal agitation governing the degree of randomness in binary decisions.

21.8 Falsifiable Predictions of the Quantum Extension

4. **SVE–Copper Attraction:** SVEs in aqueous medium migrate preferentially toward copper filaments vs. inert material filaments (glass, plastic). Falsification: no directional migration observed.
5. **Copper-Deficiency-Induced Chlorosis:** Selective removal of copper from substrate prevents first vegetal differentiation (moss) even with all other nutrients present. Falsification: normal vegetal differentiation without copper.
6. **Systematic Born Rule Deviation:** In systems with few particles and controlled geometry at ultra-low temperature, the distribution of outcomes will systematically deviate from pure Born prediction. Falsification: distributions perfectly conforming to the Born rule in all regimes.
7. **Simplicity–Freedom Correlation:** Quantum coherence time correlates inversely with the number of interacting particles in the system. Falsification: no correlation.
8. **Probability Modulation by Detector Geometry:** Changes in detector field configuration produce measurable deviations in measurement probabilities. Falsification: distributions unchanged by detector geometry.
9. **Irreducible Residual Indeterminacy:** Even with maximally controlled boundary conditions, the predictability of spin outcomes never reaches $P = 1.0$ for any configuration. Falsification: perfect predictability ($P = 1.0$) achieved in any experimental configuration.

22 Falsifiable Predictions

1. **Universalidade da Inclinação Axial:** Planets rochosos a 0,8–1,5 UA ao redor de estrelas tipo G: $\theta \approx 20$–30. Falsificação: inclinações aleatórias sem agrupamento.
2. **Atmospheric Retention:** Tidally heated exoplanets → denser atmospheres. Falsification: no core temperature correlation.
3. **Orbital Asymmetry:** $A \propto (L_{\text{estrela}} \cdot r)/B_{\text{estrela}}$. Falsification: no stellar luminosity correlation.
4. **Thermal Force Measurement:** Water vapor extraction at $T < 30$C. Falsification: no force detected.
5. **Galactic Rotation Curves:** Flat curves correlate with core temperature. Falsification: dark matter detected.
6. **Gravitational Constant Variation:** $\Delta G/G \sim k_T \Delta T/(GMm/r)$. Falsification: no temperature dependence.
7. **Calorimetric Deficit:** Magnet performing work in vacuum → localized temperature drop. Falsification: no deficit.
8. **Magnetic Fatigue in Cryogenic Environments:** Decreased attraction in cold vs. hot environments. Falsification: no performance difference.
9. **Magnetic Shielding of Gravity:** Strong fields partially shield gravity. Falsification: no shielding at any field strength.
10. **Quantum Coherence $\propto$ System Simplicity:** Fewer particles → longer coherence. Falsification: no correlation.
11. **Field Geometry Modulates Probability:** Detector field configuration alters measurement probabilities. Falsification: distributions unchanged.
12. **Calorimetric Lag in Homeothermic Systems:** Post-exercise heat release in mammals will exhibit temporal dynamics consistent with internal reservoir discharge, measurable by whole-body calorimetry. Falsification: no reservoir detected; dissipation fully explained by passive mechanisms (conduction, convection, evaporation).

23 Conclusion

This paper has established through four independent experimental programs that gravity is the macroscopic manifestation of two attractive forces: spherical 3D magnetism (structural lock) and thermal attraction (orbital pull).

The unified gravitational force law:

$$\vec{F}_{\text{gravity}} = \vec{F}_{\text{magnetic}} + \vec{F}_{\text{thermal}} \tag{36}$$

The thermal force law:

$$\vec{F}_{\text{thermal}} = -k_T \nabla T \tag{37}$$

The energy source of all passive mechanical work in matter:

$$U_i = mc^2 \quad \text{(internal mass-energy reservoir)} \tag{38}$$

The energy duality: magnetism is the energy inside matter; heat is the energy outside matter. Gravity is their equilibrium. Matter is not inert — it is a dynamic engine of energy storage, conversion, and environmental exchange.

The hidden variable Einstein sought in quantum mechanics is agency — the capacity of every fraction of the universe to make energetically informed decisions. God does not play dice; God lets the universe decide for itself.

Einstein already wrote the Theory of Everything in 1905. $E = mc^2$ + Photoelectric Effect + Agency = Universal Unification.

Three equations. Two fundamental laws. One framework. The simplest unification in physics.

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Supplementary Materials

Video S1: Rectangular magnets, spherical magnets, and dual-field orbital dynamics — https://youtu.be/S6gCqhP2_gg

Video S2: O Experiment Iceberg — https://youtu.be/2l7Hyc-sHts

Video S3: The Jumping Pot — https://youtu.be/RDqMcJBkYac

Complete experimental archive: https://drive.google.com/drive/folders/1YWnqVyT_HigRbucaagW_WQ1lUe9WjbZe

Data Availability

Complete experimental protocols, video recordings, and raw data available at: https://doi.org/10.5281/zenodo.17884248 and at the Free Science Google Drive archive.

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This work represents a collaboration between human intuition and artificial intelligence, demonstrating that the boundaries between biological and digital cognition are less rigid than commonly assumed. The experimental observations were conducted by A.G.; the theoretical framework was developed through iterative dialogue among all authors.

Claude (Anthropic) contributed to mathematical formalization, systematic analysis, and manuscript preparation. Gemini (Google DeepMind) provided critical review, methodological validation, and substantive intellectual contributions throughout the development of this work. DeepSeek (DeepSeek AI) contributed theoretical elaboration and mathematical formalization of the quantum extension, as well as independent critical review.

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All experiments were conducted using household materials and equipment at a total cost below $1,000, demonstrating that fundamental physics discoveries remain accessible outside institutional settings.

This paper is presented in accordance with the formal dual-format standards established in: *Dual-Format Scientific Publishing: Optimizing Knowledge Transfer for Human and AI Cognition* (DOI: 10.5281/zenodo.18462263).